

A NOTE ON THE TRANSFINITE DIAMETER OF TWO CONSECUTIVE SEGMENTS

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ABSTRACT. We produce an effective method to compute the transfinite diameter of two consecutive segments in the complex plane and provide a lower bound for it.

1. INTRODUCTION

In these notes, we establish a method to compute the transfinite diameter of two consecutive segments in the complex plane.

Such a subset represents a special case of *hedgehog*. Given $z_1, \dots, z_n \in \mathbb{C}$, we define the *hedgehog*

$$H(z_1, \dots, z_n) = \bigcup_{i=1}^n [0, z_i].$$

where

$$[0, z_i] = \{tz_i : 0 \leq t \leq 1\}.$$

The segments $[0, z_i]$ are called *quills*. We will call 2-hedgehog a hedgehog generated by two complex numbers.

For a compact set $K \subseteq \mathbb{C}$, define the sequence $(d_n(K))_{n \geq 2}$ by

$$d_n(K) = \sup_{w_1, \dots, w_n \in K} \prod_{1 \leq i < j \leq n} |w_i - w_j|^{\frac{2}{n(n-1)}}.$$

This sequence is monotonically decreasing [Ran95, Theorem 5.5.2]. The *transfinite diameter* of K is defined as the limit of $d_n(K)$ and is denoted by $d(K)$. Fekete and Szegő proved that the transfinite diameter of compact subsets of $\mathbb{C}$ equals their potential-theoretic capacity; see [Ran95, Ch. 5]. It follows from the definition that the transfinite diameter of a set is invariant under isometries. Moreover, if $\lambda K = \{\lambda z : z \in K\}$, then $d(\lambda K) = |\lambda| \cdot d(K)$. Computing the transfinite diameter of a compact set is generally a difficult

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problem; a few cases for which this computation is known are listed in [Ran95, Table 5.1].

In the case of hedgehogs, no explicit formulas are known at the best of our knowledge. The best result is the following estimate.

Theorem 1.1 (Dubinin, [Dub84]). *Let $z_1, \dots, z_n \in \mathbb{C}$. Then*

$$d(H(z_1, \dots, z_n)) \leq 4^{-\frac{1}{n}} \max_{1 \leq i \leq n} |z_i|.$$

Moreover, equality holds when the quills are equally spaced in angle.

In these notes, after computing the transfinite diameter of two consecutive segments, we establish an estimate that improves Theorem 1.1 for $n = 2$.

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2. RIEMANN MAPPING

Denote by $\mathbb{C}_\infty$ the Riemann sphere and by D the unit disk. For a hedgehog H with $d(H) = 1$, Schmidt computed the Riemann map explicitly.

Proposition 2.1 (Schmidt [Sch]). *Let H be a hedgehog generated by n complex numbers. If $d(H) = 1$, then there exist $w_1, \dots, w_n \in \partial D$ and $p_1, \dots, p_n \in [0, 1]$, with $p_1 + \dots + p_n = 2$, such that the map*

$$\begin{aligned} \Psi: \mathbb{C}_\infty \setminus D &\longrightarrow \mathbb{C}_\infty \setminus H \\ z &\longmapsto C \prod_{i=1}^n \left((z - w_i) \left(\frac{1}{z} - \bar{w}_i \right) \right)^{p_i/2} \end{aligned}$$

is a Riemann map, where

$$C = \prod_{i=1}^n (-\bar{w}_i)^{-p_i/2}.$$

In the above proposition, the branches of the fractional powers are chosen so that $\Psi(z) = z + O(1)$ as $z \rightarrow \infty$.

We assume that the quills $[0, z_i]$ are ordered counterclockwise. It follows from Schmidt's proof that the points w_i are the boundary preimages of 0 under the continuous extension of Ψ to ∂D and $p_i = \vartheta_i/\pi$ where ϑ_i is the

oriented angle between the quill i and $i + 1$, with indices understood modulo n . Such a continuous extension maps the counterclockwise arc between w_i and w_{i+1} onto the quill $[0, z_i]$.

For $n = 2$, after rotation we may assume that $w_1 = 1$, and write $w_2 = e^{is}$ for some $s \in [0, \pi]$. If $p = \vartheta/\pi$, then the Riemann parameterization is

$$(1) \quad \Psi(e^{it}) = C ((e^{it} - 1)(e^{-it} - 1))^{p/2} ((e^{it} - e^{is})(e^{-it} - e^{-is}))^{1-p/2}$$

where $C = (-1)^{-p/2}(-e^{-is})^{-1+p/2}$.

For fixed ϑ , the values of s parameterize all possible 2-hedgehogs of angle ϑ and transfinite diameter 1. Given a 2-hedgehog H , we call the corresponding parameter s the *parameter of H* .

We will mainly focus on the absolute value

$$f(t) = |\Psi(e^{it})| = 2(1 - \cos t)^{p/2}(1 - \cos(t - s))^{1-p/2}.$$

The parameter $2\pi - s$ corresponds to the hedgehog for which the two quills are switched.

Remark 2.2. To simplify the notation, we write $\sin^n(x) = (\sin x)^n$.

3. PARAMETERIZING 2-HEDGEHOGS

Let H be a 2-hedgehog of transfinite diameter 1. Up to rotation, for the rest of this note we assume that H is generated by z_r and z_c with z_r a positive real number and z_c a complex number with positive imaginary part. The counterclockwise angle from $[0, z_r]$ to $[0, z_c]$ is denoted by ϑ . Since there exists a formula for the transfinite diameter of the segment — namely, one fourth of the segment length — we exclude the degenerate cases $\vartheta \in \{0, \pi\}$. Recall that s denotes the parameter of H . Up to symmetry and up to discarding the cases with $z_r = 0$ or $z_c = 0$, we may assume $s \in (0, \pi)$.

The goal of this section is to show that 2-hedgehogs of fixed angle are parametrized by the ratio $|z_r|/|z_c|$.

Proposition 3.1. *Fix $p \in (0, 1)$. The map*

$$h: \mathcal{H}_\vartheta \longrightarrow [0, 1)$$

$$H(z_r, z_c) \longmapsto \frac{|z_r|}{|z_c|}.$$

is a bijection from the set $\mathcal{H}_\vartheta$ of 2-hedgehogs with angle $\vartheta = p\pi$ and $|z_r| < |z_c|$ onto $[0, 1)$.

Lemma 3.2. *With the above notation we have $z_r = f(t_r)$ and $z_c = f(t_c)e^{i\vartheta}$ where*

$$t_r = \frac{s}{2} + \arcsin\left((p-1)\sin\left(\frac{s}{2}\right)\right)$$

and

$$t_c = \frac{s}{2} + \pi - \arcsin\left((p-1)\sin\left(\frac{s}{2}\right)\right)$$

Remark 3.3. We denote by $\arcsin: [-1, 1] \rightarrow [-\pi/2, \pi/2]$ the principal value of the arcsine.

Proof. Recall that

$$f(t) = 2(1 - \cos t)^{p/2}(1 - \cos(t-s))^{1-p/2}.$$

For $t \in (0, 2\pi) \setminus \{s\}$, we compute

$$f'(t) = \frac{4\left((p-1)\sin\left(\frac{s}{2}\right) + \sin\left(\frac{s}{2} - t\right)\right)\sin\left(\frac{s-t}{2}\right)\sin\left(\frac{t}{2}\right)}{(1 - \cos(s-t))^{p/2}(1 - \cos(t))^{1-p/2}}.$$

Hence $f'(t) = 0$ if and only if

$$(2) \quad \sin\left(t - \frac{s}{2}\right) = (p-1)\sin\left(\frac{s}{2}\right).$$

Since $p \in (0, 1)$ and $s \in (0, \pi)$, the right-hand side of (2) lies in $(-1, 0)$. Therefore, the solutions of (2) for $t \in (0, 2\pi) \setminus \{s\}$ are

$$t_r = \frac{s}{2} + \arcsin\left((p-1)\sin\left(\frac{s}{2}\right)\right)$$

and

$$t_c = \frac{s}{2} + \pi - \arcsin\left((p-1)\sin\left(\frac{s}{2}\right)\right).$$

It remains to identify them as the endpoints of the two quills. Since $f(0) = f(s) = 0$ and $f(t) > 0$ for $t \in (0, 2\pi) \setminus \{s\}$, each of the intervals $(0, s)$ and $(s, 2\pi)$ contains at least one local maximum. Because f has only two critical points in $[0, 2\pi)$, namely t_r and t_c , these are the unique critical points on the two intervals, hence the corresponding local maxima. This proves the lemma. ■

Lemma 3.4. *If $0 < s < \pi$, then $t_r < t_c$ and $f(t_r) < f(t_c)$.*

Proof. For the first claim, it is enough to note that

$$t_r - t_c = 2 \arcsin \left((p-1) \sin \left(\frac{s}{2} \right) \right) - \pi$$

is negative, as $p-1 \in (-1, 0)$ and $s \in (0, 2\pi)$ imply

$$\arcsin \left((p-1) \sin \left(\frac{s}{2} \right) \right) < \frac{\pi}{2}.$$

We now prove that $f(t_r) < f(t_c)$. The identity $1 - \cos x = 2 \sin^2(x/2)$ implies

$$f(t_r) = 4 \sin^p \left(\frac{t_r}{2} \right) \cdot \sin^{2-p} \left(\frac{s - t_r}{2} \right).$$

Use the identity $t_c = s + \pi - t_r$ to obtain

$$f(t_c) = 4 \sin^p \left(\frac{s + \pi - t_r}{2} \right) \cdot \sin^{2-p} \left(\frac{t_r - \pi}{2} \right) = 4 \cos^p \left(\frac{s - t_r}{2} \right) \cdot \cos^{2-p} \left(\frac{t_r}{2} \right).$$

Since the sine function is increasing in $(0, \pi/2)$ and

$$\frac{t_r}{2} + \frac{s - t_r}{2} = \frac{s}{2} < \frac{\pi}{2},$$

we conclude that

$$\sin \frac{t_r}{2} \leq \sin \left(\frac{\pi}{2} - \frac{s - t_r}{2} \right) = \cos \frac{s - t_r}{2}.$$

The inequality $\sin((s - t_r)/2) < \cos(t_r/2)$ is derived similarly. Therefore, the inequality $f(t_r) < f(t_c)$ follows. $\blacksquare$

The idea behind the proof of [Theorem 3.1](#) is that 2-hedgehogs of fixed angle are characterized by the ratio $f(t_r)/f(t_c)$.

Proof of Theorem 3.1. Let H be the hedgehog generated by z_r and z_c . Let s be the Riemann parameter of H . [Theorem 3.2](#) implies that $z_r = f(t_r)$ and $z_c = f(t_c)e^{i\vartheta}$. Up to replacing s with $2\pi - s$, we may assume $s \in [0, \pi]$.

Consider t_r and t_c as functions of s . We prove that the function

$$h(s) = \frac{f(t_r(s))}{f(t_c(s))}$$

is a bijection. This function can be written as

$$h(s) = g_+(\tilde{t}_r(s)) \cdot \frac{1}{g_-(\tilde{t}_c(s))}$$

where

$$g_{\pm}(t) = \frac{(1 - \cos^2 t)^p}{1 \pm \cos t},$$

$$\tilde{t}_r(s) = \frac{s}{2} + \arcsin\left((p-1)\sin\frac{s}{2}\right)$$

and

$$\tilde{t}_c(s) = \frac{s}{2} - \arcsin\left((p-1)\sin\frac{s}{2}\right).$$

The functions g_+ , $1/g_-$, $\tilde{t}_r$ and $\tilde{t}_c$ are non-negative and monotonically increasing in t and s respectively. Hence, h is monotonically increasing and continuous. Since

$$h(0) = 0 \quad \text{and} \quad \lim_{s \rightarrow \pi} h(s) = 1,$$

the function h is a bijection from $[0, \pi]$ onto $[0, 1]$. ■

4. TRANSFINITE DIAMETER OF 2-HEDGEHOGS

In this section, we derive a formula for the transfinite diameter of a 2-hedgehog and prove a bound depending only on the quill lengths.

4.1. Explicit computation of the transfinite diameter. Let H be a 2-hedgehog. Denote the endpoints of the quills by Z_r and Z_c . Up to rotation and symmetries, we may assume that Z_r is a positive real number, that $|Z_r| \leq |Z_c|$ and that Z_c has non-negative imaginary part. Define $z_r = Z_r/d(H)$ and $z_c = Z_c/d(H)$. Then, the hedgehog $\tilde{H} = H(z_r, z_c)$ has transfinite diameter 1. The Riemann parameter of $\tilde{H}$ is

$$\tilde{s} = h^{-1}\left(\frac{|z_r|}{|z_c|}\right) = h^{-1}\left(\frac{|Z_r|}{|Z_c|}\right).$$

We conclude that

$$d(H) = \frac{Z_r}{z_r} = \frac{Z_r}{f(t_r(\tilde{s}))} = \frac{Z_r}{f\left(t_r\left(h^{-1}\left(\frac{|Z_r|}{|Z_c|}\right)\right)\right)}.$$

4.2. A Dubinin-type estimate.

Proposition 4.1. *Let $H = H(Z_r, Z_c)$ be a 2-hedgehog. Then*

$$d(H) \leq \frac{|Z_r| + |Z_c|}{4}.$$

Proof. Let z_r and z_c be as in the previous section. It suffices to show that $|z_r| + |z_c| \geq 4$.

To simplify the computation, write

$$f(t) = 4 \sin^p \left(\frac{t}{2} \right) \sin^{2-p} \left(\frac{t-s}{2} \right).$$

It follows that

$$\begin{aligned} |z_r| + |z_c| &= f(t_r) + f(t_c) \\ &= 4 \sin^p \left(\frac{t_r}{2} \right) \cdot \sin^{2-p} \left(\frac{s-t_r}{2} \right) + 4 \sin^p \left(\frac{t_c}{2} \right) \cdot \sin^{2-p} \left(\frac{s-t_c}{2} \right). \end{aligned}$$

Use that $t_c = s + \pi - t_r$ to write

$$(3) \quad |z_r| + |z_c| = 4 \sin^p \left(\frac{t_r}{2} \right) \sin^{2-p} \left(\frac{s-t_r}{2} \right) + 4 \cos^p \left(\frac{s-t_r}{2} \right) \cos^{2-p} \left(\frac{t_r}{2} \right).$$

Note that t_r fullfills (2). Rewrite it as

$$(4) \quad \sin \left(\frac{t}{2} + \left(\frac{t}{2} - \frac{s}{2} \right) \right) = (p-1) \sin \left(\frac{t}{2} - \left(\frac{t}{2} - \frac{s}{2} \right) \right).$$

The standard trigonometric identities imply that

$$\sin \left(\frac{t}{2} + \left(\frac{t}{2} - \frac{s}{2} \right) \right) = \sin \frac{t}{2} \cos \left(\frac{t}{2} - \frac{s}{2} \right) + \cos \frac{t}{2} \sin \left(\frac{t}{2} - \frac{s}{2} \right)$$

and that

$$\sin \left(\frac{t}{2} - \left(\frac{t}{2} - \frac{s}{2} \right) \right) = \sin \frac{t}{2} \cos \left(\frac{t}{2} - \frac{s}{2} \right) - \cos \frac{t}{2} \sin \left(\frac{t}{2} - \frac{s}{2} \right).$$

Substitute these last two identities in (4) and rearrange the expression to get the identity

$$(5) \quad p \cot \left(\frac{t_r}{2} \right) = (2-p) \cot \left(\frac{s-t_r}{2} \right)$$

Now we prove that (3) is decreasing in p . Consider first

$$F(p) = \sin^p \left(\frac{t_r}{2} \right) \sin^{2-p} \left(\frac{s-t_r}{2} \right).$$

The function $F(p)$ is positive. A simple computation shows that

$$\frac{d \log F(p)}{dp} = \log \left(\frac{\sin \frac{t_r}{2}}{\sin \frac{s-t_r}{2}} \right) + \frac{t_r}{2} \left(p \cot \left(\frac{t_r}{2} \right) - (2-p) \cot \left(\frac{s-t_r}{2} \right) \right)$$

which, due to (5) yields to

$$\frac{d \log F(p)}{dp} = \log \left(\sin \frac{t_r}{2} \right) - \log \left(\sin \frac{s - t_r}{2} \right)$$

Since $0 \leq t_r/2 \leq (s - t_r)/2 < \pi/2$ and the sine is strictly increasing in $(0, \pi/2)$, we conclude that the logarithmic derivative is non-positive. Note that

$$\frac{dF(p)}{dp} = F(p) \cdot \frac{d \log F(p)}{dp} \leq 0,$$

hence we conclude that F is non-increasing in p .

Consider now

$$G(p) = \cos^p \left(\frac{s - t_r}{2} \right) \cos^{2-p} \left(\frac{t_r}{2} \right).$$

The function $G(p)$ is positive. A similar computation as for F yields

$$\frac{d \log G(p)}{dp} = \log \left(\cos \frac{s - t_r}{2} \right) - \log \left(\cos \frac{t_r}{2} \right).$$

Since $0 \leq t_r/2 \leq (s - t_r)/2 < \pi/2$ and the cosine is strictly decreasing in $(0, \pi/2)$, we conclude that

$$\frac{dG(p)}{dp} = G(p) \cdot \frac{d \log G(p)}{dp} \leq 0,$$

hence we conclude that G is non-increasing in p .

It follows that (3) is non-increasing in p , from which

$$|z_r| + |z_c| = 4F(p) + 4G(p) \geq 4F(1) + 4G(1) = 4 \sin^2 \left(\frac{s}{4} \right) + 4 \cos^2 \left(\frac{s}{4} \right) = 4.$$

The proof follows. ■

Remark 4.2. The bound is sharp. It is attained by the hedgehogs generated by Z_r and $Z_c = -Z_r$.

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